

WASP-12b

R-1931

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_151227 · created 2026-08-06T13:31:27+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457383.77749 +/- 0.00023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.099 +/- 0.025
Transit depth (Rp/Rs)^2	0.98 +/- 0.49 [%]
Orbital Inclination (inc)	86.58 +/- 2.9
Airmass coefficient 1 (a1)	0.992 +/- 0.018
Airmass coefficient 2 (a2)	0.012 +/- 0.048
Scatter in the residuals of the lightcurve fit is	1.83 %
Best Comparison Star	#3 - [203, 189]
Optimal Method	PSF photometry
Transit Duration (day)	0.117 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.099 ± 0.025 vs 0.1178 (deviation 0.45σ)
Duration fit vs published	0.117 d vs 0.125 d (deviation 2.01σ)
Mid-time O-C	0.5 min
Depth SNR	2.4
Residual scatter	1.83 %

Light curve

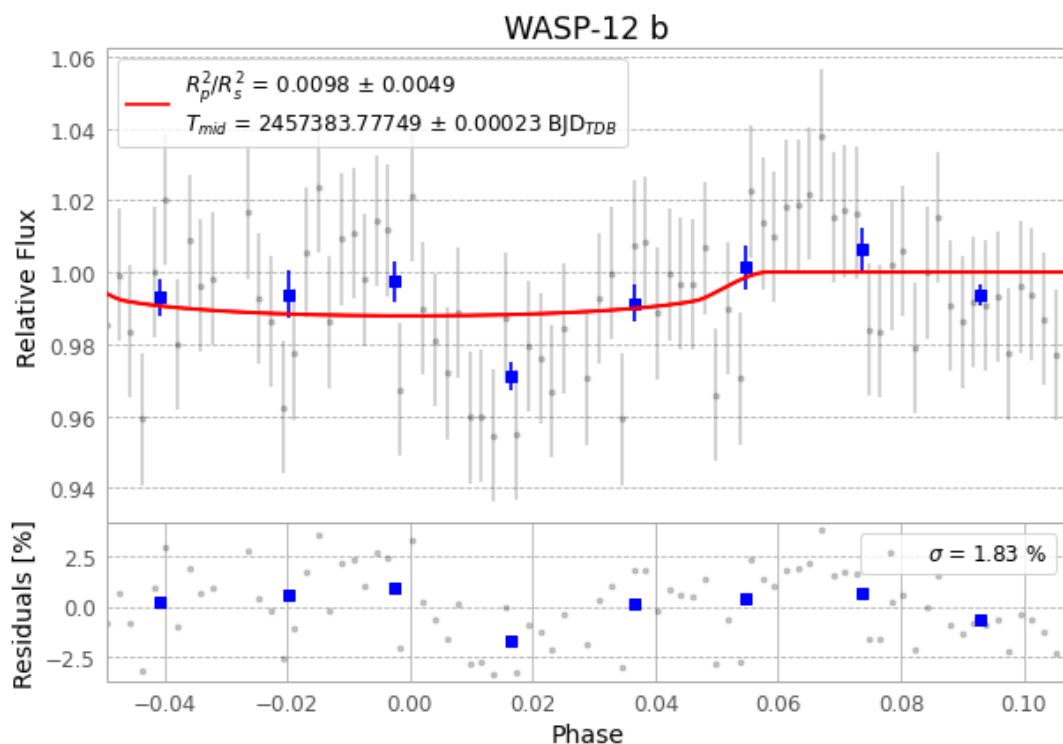


Figure 1 — FinalLightCurve_WASP-12 b_2015-12-26.png (SHA-256 2f40978bc6788b1e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [314, 205], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f184093b24cba3a6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 341f7e9

Data provenance

81 FITS frames (53 MB), WASP-12151227051212.FITS → WASP-12151227091812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-151227.log (0e81158dd24a...), FinalLightCurve_WASP-12 b_2015-12-26.png (2f40978bc678...), FinalLightCurve_WASP-12 b_2015-12-26.csv (c2f5e880f23f...)

Compute resources

Wall time 141.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 13:31Z.